

PICOFLOW · DESIGN PRECIS

PicoFlow design precis

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DOCUMENT CONTROL

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REVISION HISTORY

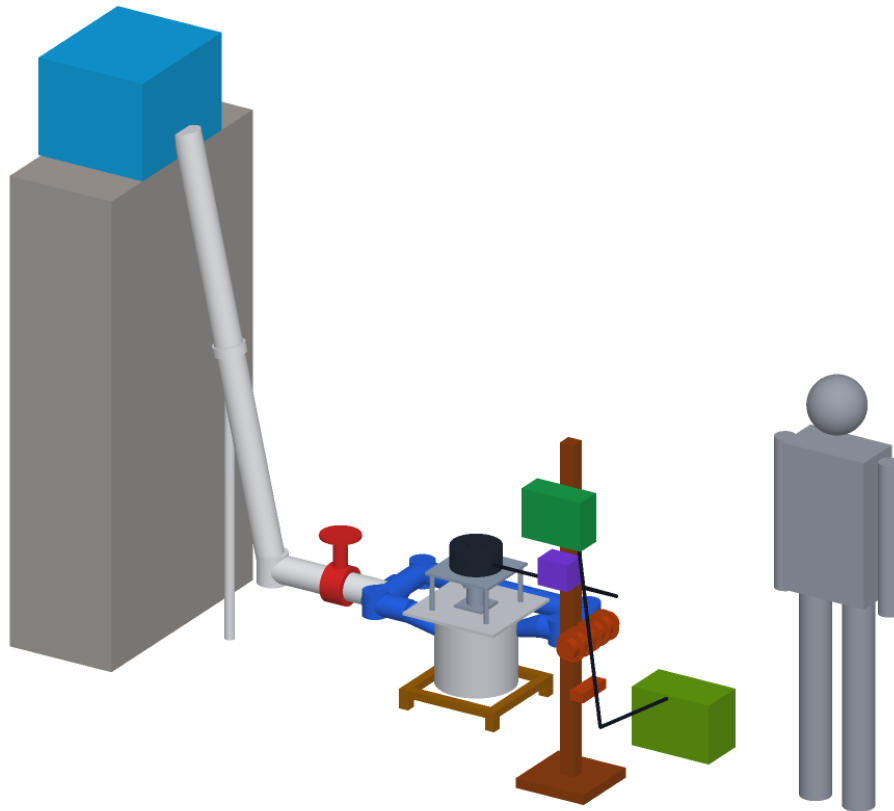
Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3 update; Amish's 2026-09-25 decisions (PCF-DDR-001), numbers from PCF-CAL-001, 90 mm branches, opposed jets, lowered frame, gate valve, voltage clamp
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, safety, media, open questions)
0.1	2026-09-24	Amish Chadha	Initial scaffold

PicoFlow design precis

PicoFlow is a vertical-shaft pico hydro turbine for 1 to 3 m of head: water from a small forebay runs down a PVC penstock to two opposed printed nozzles, which drive a 200 mm 3D-printed Turgo runner inside an open-bottomed PVC housing. The runner turns a low-speed BLDC motor, used as a generator, that sits on the lid above the spray. An MPPT controller charges a 12 V battery and diverts surplus power to a dump load, and a hardware clamp keeps the DC side below 48 V if everything else fails. The TRL 3 calculation (PCF-CAL-001) gives about 77 W into the battery at 2 m and 10 L/s (1.85 kWh a day), about 24 W at 1 m and about 146 W at 3 m. Pipe losses are about 24 % of head, more than the TRL 2 estimate, so the design point falls just short of its 80 W target; a 125 mm penstock would recover it. The turbine kit costs \$448 in parts with a new generator, against the \$450 budget Amish set on 2026-09-25, or \$378 with a salvaged washing machine motor; the penstock and battery are excluded.

PicoFlow

CONCEPT, NOT FOR FABRICATION



Grey figure: 1.75 m person for scale

Figure 1. PicoFlow at a weir or rock step set for 2.0 m of gross head, with a 1.75 m person for scale. The penstock is drawn short; real sites typically need 10 to 30 m of pipe. Kit parts are colored; the site is grey.

■ How it works

1. **Intake.** A small forebay (a plastic tub or masonry box) at the top of the drop settles sand and passes water through a 6 mm screen. Surplus water spills over the forebay and carries leaves away, so the screen cleans itself.

2. **Penstock.** A 110 mm PVC pipe carries about 10 L/s down 1 to 3 m of head. Drainage-grade pipe is allowed (decided 2026-09-25) because the static pressure is at most about 29 kPa (4.2 psi) and a multi-turn gate valve at the bottom cannot close in under about 10 s, which keeps the surge to about 35 kPa peak (PCF-CAL-001 section 9).
3. **Nozzles.** A 110 x 90 mm tee splits the flow into two 90 mm branches. One runs straight to its nozzle; the other runs around the housing, so the two jets strike the runner on opposite sides and their radial forces cancel. Each nozzle has a swappable insert (20 to 45 mm bore); the design point uses 34 mm. The jets strike the runner from above at 20 degrees to the runner plane.
4. **Runner.** A printed Turgo runner, 200 mm outside diameter and 150 mm pitch diameter, with 20 buckets, turns at about 311 rpm at the design point. The water leaves the far side of the buckets and falls out of the open bottom of the housing into the tailrace, so nothing floods the runner.
5. **Shaft and bearings.** A 20 mm stainless shaft runs up through the lid into a housing with two 6204-2RS sealed ball bearings, above the spray, and drives the generator through a guarded jaw coupling.
6. **Generator.** A new low-speed BLDC motor (about 500 W rated, about 10 rpm per volt; decided 2026-09-25) produces three-phase AC, about 27 V DC after the rectifier at the design point. Its mounting plate also accepts a salvaged direct-drive washing machine motor.
7. **Power electronics.** A three-phase bridge rectifies the output. The open-design controller (the TRL 3 design, decided 2026-09-25) runs a buck converter that holds the runner at its best speed (maximum power point tracking, MPPT) and charges the 12 V battery. When the battery is full or disconnected it switches the output into a 300 W dump load. Separately, a comparator on the DC bus switches an 8.2 Ω clamp resistor across the bus at 48 V and releases it at 40 V, whatever the microcontroller is doing.

PicoFlow: power flow at the design point, 2 m gross head and 10 L/s, W (estimates, PCF-CAL-001)

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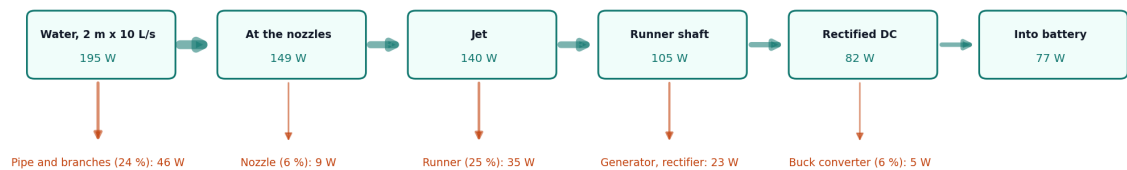


Figure 2. Power flow at the design point (2.0 m gross head, 10 L/s), from PCF-CAL-001. All values are estimates.

■ Main components

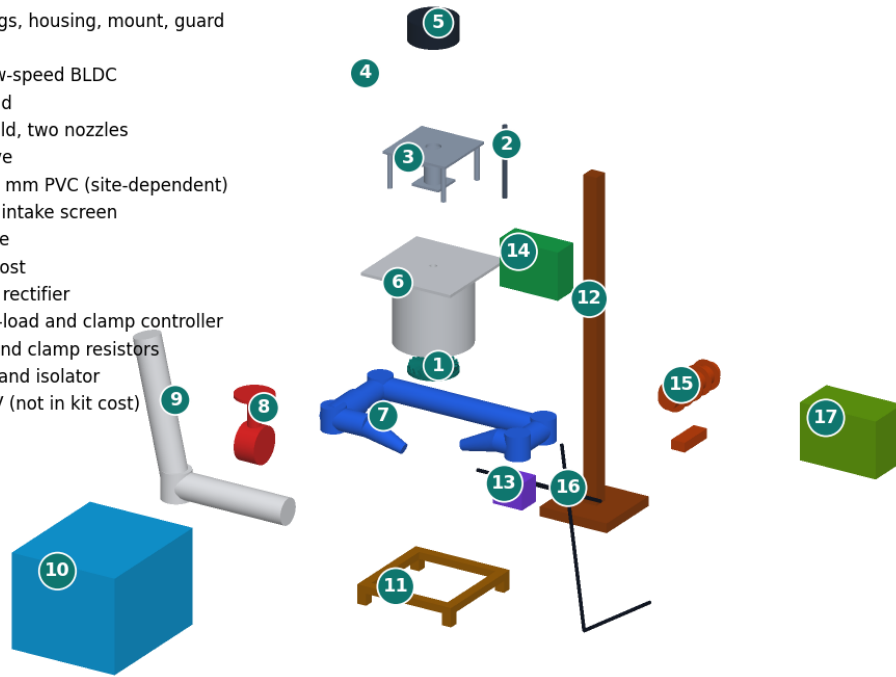
Numbers match the exploded view (Figure 3), the general arrangement drawing PCF-DWG-001 and `bom/bom.csv`.

#	Component	Choice	Notes
1	Turgo runner	200 mm OD, 150 mm pitch diameter, 20 buckets; PETG for the prototype, glass-filled nylon for field units (decided 2026-09-25)	About 525 g, about 21 h to print (R9)
2	Shaft and hub	20 mm 316 stainless shaft, 300 mm, keyed hub clamped in the runner	
3	Bearings, bearing housing, motor mount, guard	Two 6204-2RS bearings in a housing on the lid; four posts carry the generator plate; 110 mm guard around the coupling	Bearings above the spray (R11); L10 about 1.1×10^7 h
4	Coupling	Jaw coupling with elastomer spider	Inside the guard
5	Generator	New low-speed BLDC, about 500 W rated, about 10 rpm/V (decided); salvaged washing machine motor as the low-cost variant	About 7 kg
6	Housing and lid	315 mm PVC sewer pipe, 300 mm tall, open bottom at 80 mm above tailwater; 400 x 400 mm HDPE or marine plywood lid	
7	Nozzle manifold and nozzles	110 x 90 mm tee, 90 mm branches, two printed nozzles with swappable inserts, opposed jets	Inserts 20 to 45 mm (R2, R10); 34 mm at the design point
8	Inlet valve	90 mm (3 in) PVC multi-turn gate valve	Slow closing (R17)
9	Penstock	110 mm PVC, about 20 m, drainage grade allowed, support every 2 to 3 m	Site-dependent; excluded from kit cost; 125 mm proposed (see below)
10	Forebay and intake screen	Plastic tub or masonry box, 6 mm stainless mesh, overflow lip	Keep the screen clear (surge risk if a nozzle plugs)
11	Turbine frame	40 mm galvanized steel angle, 420 mm square, 80 mm legs, bolted to a pad over the tailrace	Nozzles 280 mm above tailwater (R13)
12	Equipment post	Treated timber post on a base plate	Keeps electronics out of spray and flood
13	Rectifier	Three-phase bridge, 35 A, 1,000 V, on a heat sink	
14	MPPT, dump-load and clamp controller	Microcontroller, synchronous buck converter, MOSFET dump-load switch, independent comparator clamp, IP65 box rated for 100 V DC	Open design (decided); off-the-shelf unit for first bench tests only
15	Dump-load and clamp resistors	300 W, 12 V heating element or wire-wound resistor; 8.2 Ω , 300 W aluminium-clad clamp resistor; vented guard	Both run hot
16	Wiring, fuse and isolator	4 mm ² cable, 20 A fuse at the battery, DC isolator rated 100 V, cable glands	
17	Battery	12 V (decided), about 50 Ah LiFePO ₄ with BMS, or an existing lead-acid battery	Household item; excluded from kit cost

PicoFlow: exploded view

CONCEPT, NOT FOR FABRICATION

- 1 Turgo runner, printed
- 2 Shaft and hub
- 3 Sealed bearings, housing, mount, guard
- 4 Jaw coupling
- 5 Generator, low-speed BLDC
- 6 Housing and lid
- 7 Nozzle manifold, two nozzles
- 8 Inlet gate valve
- 9 Penstock, 110 mm PVC (site-dependent)
- 10 Forebay and intake screen
- 11 Turbine frame
- 12 Equipment post
- 13 Three-phase rectifier
- 14 MPPT, dump-load and clamp controller
- 15 Dump-load and clamp resistors
- 16 Wiring, fuse and isolator
- 17 Battery, 12 V (not in kit cost)



Penstock (9) shown as a short stub; site weir omitted. Numbers match bom/bom.csv.

Figure 3. Exploded view with numbered callouts matching the BOM. The penstock is shown as a short stub and the site weir is omitted.

PicoFlow: cutaway

CONCEPT, NOT FOR FABRICATION

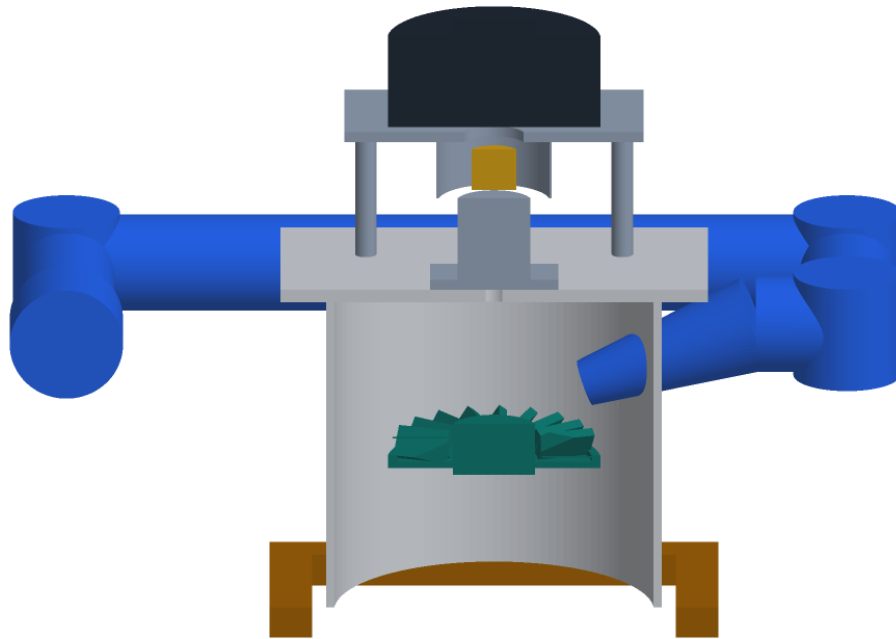


Figure 4. Cutaway of the turbine unit: opposed nozzles, runner inside the open-bottomed housing, shaft, bearing housing, guarded coupling and generator above the lid.

■ Key numbers

The numbers below are calculated in PCF-CAL-001 (script `docs/04-calcs/sizing.py`), which lists every assumption. They replace the TRL 2 first-order estimates. The general arrangement is drawing PCF-DWG-001 (`cad/drawings/PCF-DWG-001.pdf`), generated from `cad/src/model.py`.

Design point: 2.0 m gross head, 10 L/s

Quantity	Value	Basis
Gross hydraulic power	195.1 W	$\rho g Q H$
Pipe and branch loss	23.6 % of head	20 m of 110 mm PVC, fittings, 90 mm branches
Net head at the nozzles	1.53 m	
Jet velocity, jet size	5.31 m/s, two 34 mm jets	Velocity coefficient 0.97
Best runner speed, torque	311 rpm, 3.23 N·m	Bucket speed 0.46 of jet speed; runner 75 %
Rectified DC	82.1 W at 27.1 V	Generator and rectifier 78 %
Into the battery	77.2 W, 5.67 A at 13.6 V	Buck converter 94 %
Water-to-wire efficiency	39.5 %	R5 (40 %) not met
Daily energy	1.85 kWh	R6 (1.8 kWh) met

With a 125 mm penstock on the same 20 m run the loss falls to 17.5 % and output rises to 84.3 W (43.0 %), which meets R3 and R5. That change is proposed, awaiting Amish.

Across the head range

With the design-point inserts, flow rises with head. The bus stays above the 15 V the buck converter needs, so the 12 V battery charges across the whole range.

Gross head	Flow	Best speed	Bus under load	Into battery	Requirement
1.0 m	7.0 L/s	219 rpm	19.2 V	23.9 W	R4 (30 W) not met
2.0 m	9.9 L/s	311 rpm	27.1 V	77.2 W	R3 (80 W) not met
3.0 m	12.2 L/s	382 rpm	32.8 V	146.4 W	

At 1.0 m the inserts can pass at most about 10.5 L/s, so R2 (up to 15 L/s) is not met at the lowest head.

Runaway and the voltage clamp

With no load, the runner would reach about 764 rpm at 3.0 m, and the generator about 76 V DC open circuit, above the 60 V extra-low voltage limit. The hardware clamp prevents this: in the worst case (3.0 m, 15 L/s) the 8.2 Ω clamp resistor holds the runner at about 424 rpm and the bus at about 37 V, so the bus stays at or below 48 V. Only a double fault (load and clamp) lets the voltage rise to about 76 V, and the DC side is enclosed and rated for 100 V for that case. The runner rim reaches only about 7.6 m/s at runaway, so the printed runner does not burst; the generator's overspeed rating is still to be confirmed.

Setting height, loads, mass and life

- **Setting height.** The frame legs are 80 mm, so the nozzle centerline is 280 mm above normal tailwater (R13 met) and the runner underside is 210 mm above it. The setting height costs 12.3 % of the site drop at the design point, down from about 18 % at TRL 2.

- **Bearing loads** are small: with one jet closed at 3.0 m and 15 L/s, 42 N radial and 98 N axial, giving an L10 life of about 1.1×10^7 h. Seal wear and water ingress, not fatigue, will set bearing life. The printed runner's life under sandy water is unknown (R12 at risk).
- **Mass:** turbine unit (items 1 to 6 and 11) 22.2 kg, with the 7 kg generator the heaviest item; the manifold and valve add 5.2 kg and are carried separately (R14).
- **Surge:** closing the gate valve over 10 s at 3.0 m adds 0.59 m of head; an instant stop would add 44 m, which is why a quarter-turn ball valve is not allowed.

Cost

Group	Cost	Requirement
Turbine kit with a new generator (items 1 to 8, 10 to 16)	\$448.00	R15 (\$450) at risk, \$2 margin
Turbine kit with a salvaged washing machine motor	\$378.00	Documented low-cost variant
Penstock, 20 m of 110 mm (item 9, excluded)	\$100.00	Site-dependent
Battery, 12 V 50 Ah LiFePO4 (item 17, excluded)	\$160.00	Often already owned

For comparison, cheap low-head propeller units sold in Vietnam cost \$20 to \$90, but most last only 2 to 3 years (DFID R8150). PicoFlow's case rests on durability, repair and output per dollar over several years, not on first cost.

■ Key design choices

Amish decided the TRL 2 review items on 2026-09-25 by accepting every recommendation (PCF-DDR-001).

- **Turgo rather than propeller at 1 to 3 m.** Decided 2026-09-25. A Turgo runs in air, tolerates debris, keeps its efficiency at part flow, and has reached 87 % at 1 m in the lab. Its limit is flow: it suits streams of about 5 to 15 L/s (about 10 L/s at 1 m) rather than the larger flows a propeller such as the PowerSpout LH uses (14 to 55 L/s (PowerSpout)). Confirming that the first target sites are in this range is part of open item O1.
- **Vertical shaft with the generator on the lid.** Decided 2026-09-25.
- **Two jets, opposed.** Decided 2026-09-25 (two jets). Two 34 mm jets keep the jet-to-pitch ratio at 0.23; placing them on opposite sides, a TRL 3 layout choice, cancels their radial forces, and one can be closed in the dry season.
- **Generator.** Decided 2026-09-25: a new low-speed BLDC for the prototype, with a mount that also takes a salvaged direct-drive washing machine motor.
- **Runner material.** Decided 2026-09-25: PETG for the first prototype, glass-filled nylon for field units.
- **Battery voltage.** Decided 2026-09-25: 12 V.
- **Controller.** Decided 2026-09-25: the open-design MPPT, dump-load and clamp board is the TRL 3 design; an off-the-shelf diversion controller is only for the first bench tests (TRL 4, on hold).
- **Drainage pipe for the penstock.** Decided 2026-09-25: allowed with a slow-closing valve and a surge check. The check is in PCF-CAL-001 and the multi-turn gate valve meets R17.
- **Budget.** Decided 2026-09-25: \$450, with the salvaged-motor variant documented.
- **90 mm branches.** A TRL 3 sizing result: 63 mm branches would lose 33 % of the head instead of 24 %.

Proposed, awaiting Amish (raised at TRL 3): a 125 mm design-point penstock (N1), relaxing R4 to 20 W (N2), stating R2 as 5 to 10 L/s at 1.0 m (N3), confirming the R14 definition (N4) and accepting the \$2 cost margin (N5). Details are in `docs/REVIEW.md`.

■ Safety

Safety: PicoFlow combines moving water at a weir, a spinning runner, a generator that can exceed 60 V DC if both the load and the clamp fail, resistors that run hot, and a lithium battery. Treat each as a hazard at every stage, including site survey.

- **Water and the site.** Weirs, rock steps and streams in flood can drown people, especially children. Install and service only at low flow, never stand on a weir crest, keep the intake and tailrace fenced or covered where children play, and site the turbine above normal flood level. Close the intake before entering the stream. The runner is only 210 mm above normal tailwater, so check flood levels before siting.
- **Rotating parts.** The runner is enclosed in the housing, and the coupling now has a guard. Always close the valve and wait for the runner to stop before opening the housing or removing the guard; a runner turning at 300 to 400 rpm can still cut fingers.
- **Runaway voltage.** The hardware clamp keeps the DC bus at 48 V or less, but a double fault can reach about 76 V DC at 3 m head. The DC side must stay enclosed and rated for at least 100 V.
- **Dump load and clamp heat.** The 300 W dump load and the 300 W clamp resistor can exceed 200 °C in still air. Mount them in a vented metal guard, away from timber, dry grass and roofs, or immerse the dump load in a water tank with a thermal cut-out.
- **Lithium battery.** A LiFePO₄ battery is less prone to thermal runaway than other lithium chemistries but can still deliver hundreds of amperes into a short circuit. Use a battery with a BMS, a 20 A fuse within 300 mm of the battery terminal, charge-temperature limits from the BMS, and a dry, ventilated, non-combustible location. Lead-acid batteries vent hydrogen while charging and need ventilation.
- **Water hammer.** Use only the multi-turn gate valve and close it over at least 10 s. Never fit a quarter-turn ball valve: an instant stop on a 20 m penstock could add about 44 m of head and burst drainage pipe. Keep the screen clear, because a nozzle plugged all at once could add about 22 m.
- **Environment and permits.** Leave enough water in the stream for fish and downstream users (R16), screen the intake, and check local water-use rules before building.

■ Open questions (TRL 4 is on hold)

- Measure the chosen generator's constant (rpm per volt), resistance, losses at 200 to 400 rpm and overspeed rating.
- Confirm the runner efficiency of a printable bucket shape against the Bristol low-head results.
- Confirm the drainage pipe joint rating with the supplier.
- Check PETG and nylon creep, water uptake and sand erosion under continuous duty (R12).
- Choose the first site type, region and partner (O1).

Concept media: [blueprint sheet](#), [interactive 3D model](#).